Aayush Mishra

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Education

University of Birmingham, MS in Advanced Computer Science — Sept 2024 – Sept 2025

- Coursework: Intelligent Systems, Complexity and Design of Algorithms, Advanced Networking, Game Theory
- Additional Focus: Distributed Systems, Scalable Architectures, Optimization (Linear & Nonlinear Programming)

UIET, Maharishi Dayanand University, Rohtak, BTech in Computer Science — Oct 2020 – June 2024

- Coursework: Operating Systems, Data Structures, Theory of Computation

Experience

AI Engineer, Zenalyst.ai – Bengaluru, IN — Jun 2026 – Present

- Cut agent context 30× (~93K → ~3.1K tokens) with 100% recall on a 28-question benchmark over 9.9M rows by building ZenContext, a retrieval layer combining a compact Data Map, hybrid BM25 + vector search and sub-ms lookups, served to agents over MCP.
- Shipped 24 production agents with zero new runtime code by architecting a manifest-driven multi-agent framework (LLM planner + ReAct runtime, Python/FastAPI) with parallel sub-agent fan-out, where routing, prompts, model tier and 31 tools live in one YAML file.
- Brought in-call voice response latency down to 300–400 ms for 8 production outbound voice agents (incl. autonomous Collections) by building a streaming speech pipeline (VAD → STT → LLM → TTS) on LiveKit with Sarvam/Deepgram, Groq Llama-3.3-70B, barge-in and mid-call tool calls over SIP in English/Hindi.
- Delivered a Legal AI workbench (contract review, bilingual title due diligence, multimodal deed reconciliation) by grounding every LLM output in server-validated citations and pairing LLM extraction with deterministic red-flag rules.
- Blocked hallucinated financial figures and cross-tenant leakage by building an LLM-as-a-judge eval harness (RAGAS-style faithfulness, gated at 0.90 with regression evals in CI) and enforcing RLS-scoped read-only agent tokens, SQL guardrails and PII masking, backed by 400+ test suites.

Software Engineer – AI, Aeoru – Delhi, IN — Jan 2026 – May 2026

- Increased ANPR accuracy by 20% on real-world telemetry by retraining plate-recognition models on domain data and adding VLM-based verification for low-visibility frames.
- Deployed in-bus camera analytics on 50+ buses (headcount, drowsiness, cabin events) with real-time NPU inference on Radxa/NanoPi edge devices by owning data curation, CV model fine-tuning and edge deployment.
- Improved alert quality in multi-camera workflows by fine-tuning LLM/VLM components for event interpretation on an object-detection and human-behavior analysis platform.

Projects

Ryvos — Github Repo

- Achieved ~15MB idle RAM and <50ms task routing in a 40k+ LOC Rust autonomous AI agent by parallelizing tool execution on the Tokio async runtime.
- Secured LLM tool execution against prompt injection via a 5-tier SecurityGate (Lua/Rhai sandboxing, dangerous-command heuristics, human-in-the-loop), while integrating 16+ LLM providers and 50+ tools across 6 channels.

Skills

Languages: Python, Rust, C++, Java, Go, SQL, TypeScript

Technologies: FastAPI, Spring Boot, PostgreSQL, Redis, MongoDB, ClickHouse, PySpark, Docker, AWS (SageMaker, Bedrock), GCP, Azure

AI/ML: PyTorch, Speech (STT/TTS, LiveKit), CUDA, LLM Agents, RAG, LLM Evals, MCP, LangGraph, Fine-Tuning, VLMs, OpenCV, ONNX, TFLite, Edge Inference, Model Quantization